

Hassan Elhllos

Embedded Systems & PCB Design Engineer

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PROFESSIONAL SUMMARY

Embedded Systems and PCB Design Engineer with 3+ years of freelance experience building custom IoT hardware end-to-end, from schematic design and PCB layout to firmware development (C/C++) and cloud integration. Proficient with ESP32, STM32, KiCad and Altium Designer, with hands-on industrial automation (Siemens S7-1200, Schneider PLC, VFD, Modbus RS485). Won 1st place at Suez Canal University's startup competition for an industrial PLC product.

TECHNICAL SKILLS

Embedded Systems: ESP32, STM32, ESP12E/F, Arduino, Raspberry Pi, GPIO, Timers, Interrupts

PCB and Hardware Design: KiCad, Altium Designer, Schematic Design, PCB Routing, EMC-hardened layout, Proteus, Soldering (through-hole and SMD)

Industrial Automation: Siemens S7-1200 PLC, Schneider PLC, Ladder Logic, Modbus RTU (RS485), VFD configuration, Classic Control (star-delta, overload protection)

Software and Data: C/C++ firmware, Python, SQL, Git/GitHub, JSON, Ladder Diagram, Dashboard Development

Connectivity: MQTT (HiveMQ), UART, SPI, I2C, RS485, HTTP, ESP RainMaker, Blynk, Wi-Fi, BLE

Computer Vision and Lab: OpenCV, Object Detection, Raspberry Pi, Oscilloscope, Multimeter, EMI troubleshooting

EXPERIENCE

Independent IoT and Hardware Developer
Freelance

2023 – Present

- Design custom IoT prototypes and PCB layouts for academic and client projects, with cloud platforms (ESP RainMaker, Blynk) for remote control and rapid deployment.

Industrial Automation Trainee
HA Consulting Group

Sep 2025 – Nov 2025

- Configured VFDs and designed classic control circuits (star-delta starter, overload protection); programmed Siemens S7-1200 and Schneider PLCs.

IoT Intern
Creativa Innovation Hub (ITIDA), Grade: Excellent

May 2025 – Jun 2025

- Configured ESP32 edge platforms with MQTT (HiveMQ); built cloud dashboards and an end-to-end Smart Agriculture system with edge-to-cloud communication.

Computer Vision Volunteer
SCU Racing Team (ROV), Suez Canal University

2024 – 2025

- Developed underwater object-detection and navigation modules with Python and OpenCV on Raspberry Pi.

Data Analytics Intern
Information Technology Institute (ITI)

Jul 2024 – Aug 2024

- Used SQL and data-cleaning to extract insights, then built dashboards for real-time clarity.

SELECTED PROJECTS

Industrial PLC Controller: custom PCB hardened for EMC, 8 I/O, Modbus to HMI and SCADA, in-house Ladder-programming IDE, 3D-printed enclosure; 1st place at Canal Startup Sprint.

Custom Smart Home System: ESP32 with 8-channel relays on a custom KiCad PCB; remote access, schedules and scene control via Blynk/ESP RainMaker.

ESP32 DevKit Board: KiCad board with auto-programming (one-wire) USB-UART, removing the BOOT button requirement.

Star-Delta Starter Control Panel: contactors, timer and overload relay for safe three-phase induction-motor starting.

Cold Storage Monitoring and Control: dual ESP32 boards read temperature and fault lamps of industrial cold-storage panels with remote alerts.

EDUCATION

B.Sc. in Electrical Engineering
Suez Canal University, Faculty of Engineering

2021 – 2026

Diploma in Information Technology
Advanced Technical School - Ismailia, Egypt

2016 – 2021

Advanced Industrial Electronics Diploma
Digital Fabrication - High-Speed PCB Design and SMPS

Ongoing

HONORS AND CERTIFICATIONS

1st Place — Canal Startup Sprint at Suez Canal University (graduation project: Industrial Controller ESP32 PLC).

Certifications: Industrial Automation (HA Consulting), IoT Internship Excellence and Excellent grade (ITIDA), Data Analysis (ITI).

LANGUAGES

Arabic (Native) and English (Professional)